

RATIO & PROPORTION

By:-

**Naveen Kumar Jain
(Aptitude Trainer)**



RATIO

- ❑ A ratio is a comparison of two quantities by division.
- ❑ It means what part one quantity is of another.
- ❑ If **a** and **b** are two numbers, then the ratio of **a** to **b** is $\frac{a}{b}$ or $a \div b$
It is denoted by **a:b**.

In the ratio **a:b** **a** is called first term or **antecedent**

b is called second term or **consequent**

**In a certain class, the ratio of passing grades to failing grades is 7 to 5.
How many of the 36 students failed the course?**

The ratio, "7 to 5", (or $7 : 5$ or $7/5$) tells you that, of every $7 + 5 = 12$ students

Out of 12 students 5 failed.

That is, $5/12$ of the class flunked.

Then $(5/12)(36) = 15$ students failed.

TYPES OF RATIOS

Duplicate Ratio: The ratio of squares of two numbers is called the duplicate ratio of the two numbers.

Example : The duplicate ratio of $2/5 = 4/25$

Triplicate Ratio: The ratio of cubes of two numbers is called the triplicate ratio of the two numbers.

Example: The triplicate ratio of $2/5 = 8/125$

Sub-duplicate Ratio: The ratio of the square root of two numbers is called the sub-duplicate ratio of the two numbers.

Example: The sub-duplicate ratio of $4/25 = 2/5$

Sub-triplicate Ratio: The ratio of the cube root of two numbers is called the sub-triplicate ratio of the two numbers.

Example The sub-triplicate ratio of $8/125 = 2/5$

TYPES OF RATIOS

Inverse Ratio or Reciprocal Ratio: If the antecedent and consequent of a ratio interchange their places, the new ratio is called the inverse ratio of the first.

If $a : b$ be the given ratio, then $1/a : 1/b$ is its inverse ratio.

Example: The inverse ratio of $3/4 = 4/3$

Compounded Ratio: The compounded ratio of the ratios: $a : b :: c : d$
$$= \frac{ac}{bd}$$

PROPORTION

- ❑ The equality of two ratios is called Proportion.

If $\mathbf{a/b=c/d}$, then $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ and $\mathbf{d}$ are said to be in proportion

It is written as $\mathbf{a:b :: c:d}$

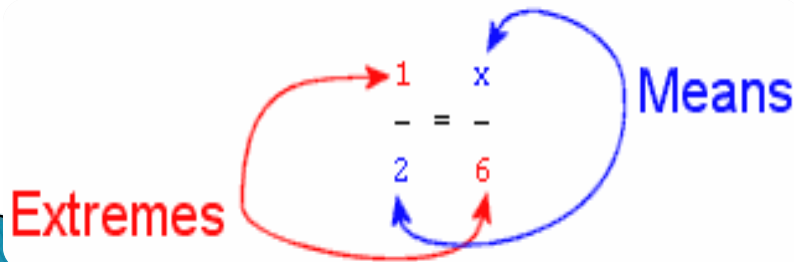
- ❑ Each term of the ratio a/b and c/d is called a proportion a , b , c and d are respectively the first, second, third and the fourth proportional.

- ❑ Here a , d are known as extremes.
 b , c are known as means.

Product of means = Product of extremes.

Thus $a : b :: c : d$

$$\Rightarrow a \times d = b \times c$$



BASIC TERMS

Mean Proportional: If $a:x :: x:b$,
 x is called the mean or second proportional of a . b .

$$\begin{aligned}\text{We have, } \frac{a}{x} &= \frac{x}{b} \\ \text{or } x^2 &= ab \\ x &= \sqrt{ab}\end{aligned}$$

Thus, Mean proportional of **a** and **b** is $\sqrt{ab}$.

We also say that **a**, **x**, **b** are in continued proportion.

Q. Find the mean proportional between 4 and 64.

$$\text{Solution: } x^2 = 4 \times 64$$

$$x = 16$$

BASIC TERMS

Fourth Proportion: If $a:b::c:x$,

$$x = \frac{bc}{a} \quad x \text{ is called the fourth proportional of } a, b \text{ and } c.$$

Find the fourth proportion to the numbers 4, 10, and 12.

Sol: Fourth Proportion $x = \frac{10 \times 12}{4} \rightarrow x = 30$

Third Proportion: For $a:b, x$ is called the third proportion.

$$\text{If } a:b :: b:x$$

$$x = \frac{b^2}{a}$$

Find the third proportion to the numbers 4, 12.

Sol: Third Proportion $x = \frac{12 \times 12}{4} \rightarrow x = 36$

PRACTICE QUESTIONS

Q.2 If $A : B = 5 : 9$ & $B : C = 4 : 7$, find $A : B : C$.

Given $A : B = 5 : 9$ and $B : C = 4 : 7$

$A : B : C$

$5 : 9$ $\times 4$

$4 : 7$ $\times 9$

Make the common value same by
multiplying both equation

$20 : 36$

$36 : 63$

Therefore $A:B:C = \mathbf{20:36:63}$

3. Find the third proportional to 16 and 36.

A. 24

B. 36

C. 64

D. 81



4. Physics tells us that the weights of objects on the moon are proportional to their weights on earth. Suppose a 180 lb. man weighs 30 lb. on the moon. What is the weight of a boy who weighs 15 lb. on the moon?
- A. 2.5 lb. B. 5.0 lb. C. 45 lb. D. 90 lb.
- 

5. If $2A = 3B = 4C$, then $A : B : C$ is

A. 2:3:4

B. 4:3:2

C. 3:6:4

D. 6:4:3

6. F is directly proportional to A. If $F = 42$ when $A = 6$, then find the value of A when $F = 140$.

A. 14.5

B. 20.0

C. 25.2

D. 26.4

7. The amount of Rs. 540 is divided among three members A, B and C in the ratio of 1: 2: 3. Find the share of B.
- A. Rs. 180 B. Rs. 100 C. Rs. 270 D. Rs. 90

8. A certain amount was divided among three persons P, Q & R in the ratio of 9: 12: 16. P and Q together got Rs. 8379. Find the amount that R got.
- A. Rs. 6584 B. Rs. 6834 C. Rs. 6384 D. Rs. 6484

9. Find the value of $(4x + 3y) : (6x + 7y)$, if $x : y = 1 : 4$

A. 6: 11

B. 8: 17

C. 5: 13

D. 4: 5

10. If $x: y = 3: 2$ and $y: z = 5: 7$, find $x: y: z$.
- A. 3:10:7 B. 15: 10:21 C. 10:15:14 D. 15:10:14

Q.11 Gold and Brass are in ratio of 12:7. If the mixture weight is 38.19 grams, then the weight of brass is?

- A. 13.23 grams B. 14.07 grams C. 15.08 grams D. 12.16 grams**

Q.12 Length and breadth of a rectangle are in the ratio of 9 : 7. The circumference of the rectangle is 480 meters. Then the length of the rectangle?

- A. 136 m B. 135 m C. 132 m D. 140 m**
- 

13 **A man earned ₹13.6 million. He distributed the money to his wife, son and daughter in the ratio of $5/2:7/2:2$. Then find the share of his son?**
A. 5.95 million B. 6.85 million C. 7.65 million D. 6.95 million

14. **When distributed the amount Rs. 125000 to A, B and C in the ratio 9 : 10 : 6, find the amount that B gets more than C?**
A. Rs. 20,000 B. Rs. 22,000 C. Rs. 18,000 D. Rs. 24,000



15 If $x:y = 4:5$, then the ratio of $2x+3y$ and $3x-2y$?
A. 24:5 B. 23:4 C. 23:2 D. 25:6

16 The ratio of the marks of Ravi and Krishna is $7 : 8$ and that of Krishna and Harish is $5 : 6$. The marks obtained by Ravi is 350. Find the marks of Harish?
A. 460 B. 480 C. 400 D. 450



- 17 The ratio of two numbers is 5 : 6. If 10 is added to both the numbers, the ratio of the numbers is 7 : 8, then find the numbers?
- A. 30, 36 B. 20, 24 C. 25, 30 D. 30, 35

18. In an office, they reduced the employees in the ratio of 25:18. They increased their wages in the ratio of 9:14. Then find the ratio of wages bill of previous salaries and present salaries of the employees?
- A. 25:28 B. 22:28 C. 25:18 D. 18:25
- 

- 19 Monthly income of Virat and Rohit are in the ratio of 11 : 9. Their expenses are in the ratio of 7 : 5. They save ₹ 5000 each. Then find the monthly income of Rohit?
- A. Rs. 11,650 B. Rs. 11,550 C. Rs. 12,750 D. Rs. 11,250
- 20 The investments of A, B and C in business are ₹20000, ₹30000, ₹36000 respectively. They worked for the time period of 12, 8 and 9 months respectively. Their ratio of shares of business in the profit is?
- A. 15:16:18 B. 24:20:27 C. 20:24:27 D. 20:20:27
- 

PARTNERSHIP

- ❑ To run business or any organization, we need to have partnerships.
- ❑ There are two types of partners:

Working Partner: Those partners who invest the money and manage the business or affairs of organization are called working partners.

Sleeping Partners: Those partners who merely invest money and do not involve in business affairs are called sleeping partners.

Case 1: When time period of investment is constant, then profit of partners will be divided into ratio $C_1:C_2:C_3.....:C$

Case 2: When amount invested is same for different partners but time period is different, then profit of partners will be divided into the ratio of their time invested $t_1:t_2:t_3.....:t_n$

Case 3: When amount of investment and time period both are different for different partners, then profit will be divided into following ratio. $C_1t_1:C_2t_2:C_3t_3.....:C_nt_n$

where C_1, C_2, C_n are the respective amounts and $t_1:t_2:t_3.....:t_n$ are the respective time period of their investment.

PARTNERSHIP

- 1. Three partners shared the profit in a business in the ratio 5 : 7 : 8. They had partnered for 14 months, 8 months and 7 months respectively. What was the ratio of their investments?**

A. 5 : 7 : 8 B. 20: 49: 64 C. 38: 28: 21 D. None of these

- 2. A starts business with Rs. 3500 and after 5 months, B joins with A as his partner. After a year, the profit is divided in the ratio 2 : 3. What is B's contribution in the capital?**

A. Rs. 750 B. Rs. 850 C. Rs. 800 D. Rs. 900

3. Manoj got ₹6000 as his share out of a total profit of ₹9000 which himself and Ramesh got at the end of one year. If Manoj invested ₹20000 for 6 months, whereas Ramesh invested his amount for the whole year, what was the amount invested by Ramesh?
- A. ₹7000 B. ₹5000 C. ₹5500 D. ₹6000
4. Dilip, Ram and Amar started a shop by investing ₹27000, ₹81000 and ₹72000 respectively. At the end of one year, the profit was distributed. If Ram's share of profit be ₹36000, the total profit was?
- A. ₹80,000 B. ₹70,000 C. ₹75,000 D. ₹60,000
- 

- 5, A and B started a joint firm. A's investment was thrice the investment of B and the period of his investment was two times the period of investment of B. If B got ₹4000 as profit, then their total profit is?
A. ₹24,000 B. ₹30,000 C. ₹28,000 D. ₹32,000
6. Jayant started a business, investing ₹6000. Six months later Madhu joined him, investing ₹4000. If they made a profit of ₹5200 at the end of the year, how much must be the share of Madhu?
A. ₹1280 B. ₹1250 C. ₹1300 D. ₹1400
- 

7. ₹700 is divided among A, B, C so that A receives half as much as B and B half as much as C. Then C's share is?
A. ₹200 B. ₹600 C. ₹300 D. ₹400
8. A, B and C invest ₹2000, ₹3000 and ₹4000 in a business. After one year, A removed his money but B and C continued for one more year. If the net profit after 2 years be ₹3200, then A's share in the profit is?
A. ₹1000 B. ₹600 C. ₹400 D. ₹800
- 

9. Suresh and Rohit invest in a business in the ratio of 7:3. If they decide to donate 20% of profit to charity and Rohit's share is ₹490, then the total profit is?
A. ₹2041 B. ₹1999 C. ₹2021 D. ₹1998
10. Three people A, B and C invest ₹100000 for a business. B invest ₹10000 more than A and C invest ₹5000 more than B. Out of a total profit of ₹50000. What is the share of A?
A. ₹11750 B. ₹12250 C. ₹12500 D. ₹13225
- 

11. A business yields a profit of ₹36000 over a year. The capital share of A and B is same, but B joined business 4 months later. Find the share of A in the profit.
A. ₹21600 B. ₹14400 C. ₹18900 D. None of these
12. Ajay and Bala started a partnership business by investing capital in the ratio of 5:3. Chandra joined them after 8 months with an amount equal to that of Ajay. In what proportion should the profit be distributed among them at the end of the year?
A. 1: 2: 3 B. 15: 9: 5 C. 5: 15: 9 D. 14: 11: 9
- 

13. By renting a pasture, A puts 9 oxen for 3 months and B puts 2 oxen for 9 months. If the rent of the pasture is ₹150, what is A's share in the rent?
A. ₹90 B. ₹60 C. ₹100 D. ₹50
14. Sachin started a dairy business by investing ₹25000. After 6 months, Nilesh joined him with the capital of ₹40000. After 2 years they earned a profit of ₹50000. What was Sachin's share in that profit?
A. ₹21525 B. ₹22727 C. ₹23494 D. ₹24588
- 

15. Ganesh, Sai and Krishna invested ₹5000, ₹10000 and ₹5000 respectively. Sai left after 5 months. If after 10 months, the gain was ₹9000, then what will be the share of Krishna?
- A. ₹1000 B. ₹2000 C. ₹3000 D. ₹4500
16. A invested 100% capital in business. The business was run by B for 4 months and by C for 6 months. It was agreed that A will take 50% of profit. If the profit was ₹40000, then what is the share of C in that profit?
- A. ₹11000 B. ₹13000 C. ₹14000 D. ₹12000
- 

23. The ratio of the ages of Minu and Meera is 4:2. If the sum of their ages is 6 years, find the ratio of their ages after 8 years.

A. 8:6

B. 6:5

C. 6:4

D. 7:5

24. The ratio of the ages of Seeta and Geeta is 2:7. After 6 years, the ratio of their ages will be 1:2. What is the difference in their present ages?

A. 8 years

B. 9 years

C. 10 years

D. 11 years



- 25** **Ten years ago, the sum of ages of a father and his son was 34 years. If the ratio of present ages of the father and son is 7:2, find the present age of the son.**
- A. 12 years B. 11 years C. 10 years D. 8 years**

- 26.** **The sum of the ages of father and his son is 44 years. If 6 years after the father will be 3 times as old as his son, what are their present ages?**
- A. 38, 8 B. 38, 6 C. 35, 9 D. 37, 7**
- 

27 **Ten years ago, Ajay was $\frac{1}{3}$ rd as old as Vijay. If Ajay is 18 years old now, how old is Vijay now?**

- A. 32 years B. 34 years C. 36 years D. 38 years**

28. **Age of father is seven times the age of his son. In the next 10 years, the age of father will be three times the age of his son. What is the age of father now?**

- A. 35 years B. 37 years C. 39 years D. 41 years**
- 

29. Twelve years ago, Budh was twice as old as Badri. If the ratio of their present ages is 4:3 respectively, find the difference between their present ages.
A. 5 years B. 6 years C. 7 years D. 8 years
30. Two years ago the ratio of ages of Naresh and Suresh was 3:2. If one year hence, the ratio of their ages will be 7:5, what is the sum of their present ages?
A. 30 years B. 32 years C. 34 years D. 36 years
- 

- 31** The sum of the present age of mother and her son is 60 years. Six years ago, the age of the mother was five times the age of her son. What will be the age of her son after 6 years?
- A. 23 years B. 22 years C. 20 years D. 21 years

- 32** Simran is younger than Simarjeet by 7 years, and the ratio of their ages is 7:9. What is the age of Simran?
- A. 24 years B. 23.5 years C. 24.5 years D. 25 years
- 

33. My brother, Ravi is 3 years elder to me. My father's age was 28 years when my sister, Neha was born, and my mother's age was 26 years when I was born. If Neha's age was 4 years when Ravi was born, what was the age of my father when Ravi was born?
- A. 35 years B. 34 years C. 33 years D. 32 years
34. The product of the ages of Shyam and Ram is 240. If double the age of Ram is more than Shyam's age by 4 years, what is Ram's present age?
- A. 16 years B. 14 years C. 12 years D. 10 years
- 

35

The present ages of Arun, Sunil, and Vivek are in proportions 4:7:9. Before eight years, the sum of their ages was 56. What are their present ages in years?
A. 16, 30, 40 B. 16, 28, 40 C. 16, 28, 36 D. Insufficient data

36

Sobha's father was 38 years of age when she was born while her mother was 36 years old when her brother four years younger to her was born. What is the difference between the ages of Shoba's parents?
A. 6 years B. 5 years C. 4 years D. 3 years

- 37** Vipul's present age is two-fifths of the age of his mother. After 8 years, Vipul will be one-half of the age of his mother. What is the present age of Vipul's mother?
- A. 60 years B. 50 years C. 40 years D. 30 years**

- 38** The ages of Ram and Shyam differ by 16 years. 6 years ago, the Ram was 3 times as old as Shyam. What is the present age of Ram?
- A. 40 years B. 30 years C. 20 years D. 15 years**
- 

39

If 6 years are subtracted from the present age of Vijay and the remainder is divided by 18, the present age of Vishal is obtained. If Vishal is 2 years younger to Denis whose age is 5 years, then what is Vijay's present age?

A. 50 years

B. 60 years

C. 65 years

D. 62 years

40

Father is 3 times more aged than his daughter. If after 5 years, he would be 3 times of daughter's age, then further after 5 years, how many times he would be of his daughter's age?

A. 2 times

B. 2.5 times

C. 3 times

D. 3.5 times

